

ACTION POTENTIAL

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- **Definition :** A series of electrical changes that occur in membrane potential when muscle / nerve is stimulated.
- **Action potential occurs in 2 phases :—**
 - a. **Depolarisation :** Initial phase of action potential in which inside becomes positive & outside becomes negative. Polarized state (RMP) is abolished, resulting in depolarisation.
 - b. **Repolarisation :** Phase of action potential which reverses back to RMP (inside becomes negative & outside becomes positive). Polarised state is re-established.

PROPERTIES OF ACTION POTENTIAL VS GRADED POTENTIAL

Action Potential	Graded Potential
• Propagative	• Non-propagative
• Long-distance signal	• Short-distance signal
• Both depolarization and repolarization	• Only depolarization or hyperpolarization
• Obeys all-or-none law	• Does not obey all-or-none law
• Summation is not possible	• Summation is possible
• Has refractory period	• No refractory period

ACTION POTENTIAL CURVE

- **Definition** : Graphical registration of electrical activity that occurs in an excitable tissue (muscle after excitation / stimulation).
- **3 Major Parts** :—
 - a. Latent period
 - b. Depolarisation
 - c. Repolarisation

1. Latent Period

- Period when no change occurs in electrical potential immediately after applying stimulus.
- **Duration** : 0.5 – 1 msec.

Stimulus Artifact :—

When stimulus applied → slight irregular deflection of baseline for very short period → because of disturbance in muscle due to leakage of current from stimulating electrode to recording electrode.

2. Depolarization

- Muscle depolarized for about 15 mV (up to -75 mV).
- **Firing Level** : After initial slow depolarisation for 15 mV → rate of depolarisation increases suddenly → point at which depolarisation rate increases is the **firing level**.
- **Overshoot** : From firing level → curve reaches isoelectric potential (0 potential) → then shoots up (overshoots) beyond 0 potential up to **+55 mV**.

3. Repolarisation

- Initially, repolarization occurs rapidly and then becomes slow.
- **Spike Potential** : Rapid rise in depolarisation & rapid fall in repolarisation taken together.
 - **Duration** : 0.4 msec.

- **After-Depolarisation / Negative After-Potential** : Rapid fall in repolarisation followed by a slow repolarisation.
 - **Duration** : 2 – 4 msec.
- **After-Hyperpolarisation / Positive After-Potential** : After reaching resting level (-90 mV) → becomes more negative beyond resting level → after this, RMP is restored.
 - **Duration** : 50 msec.

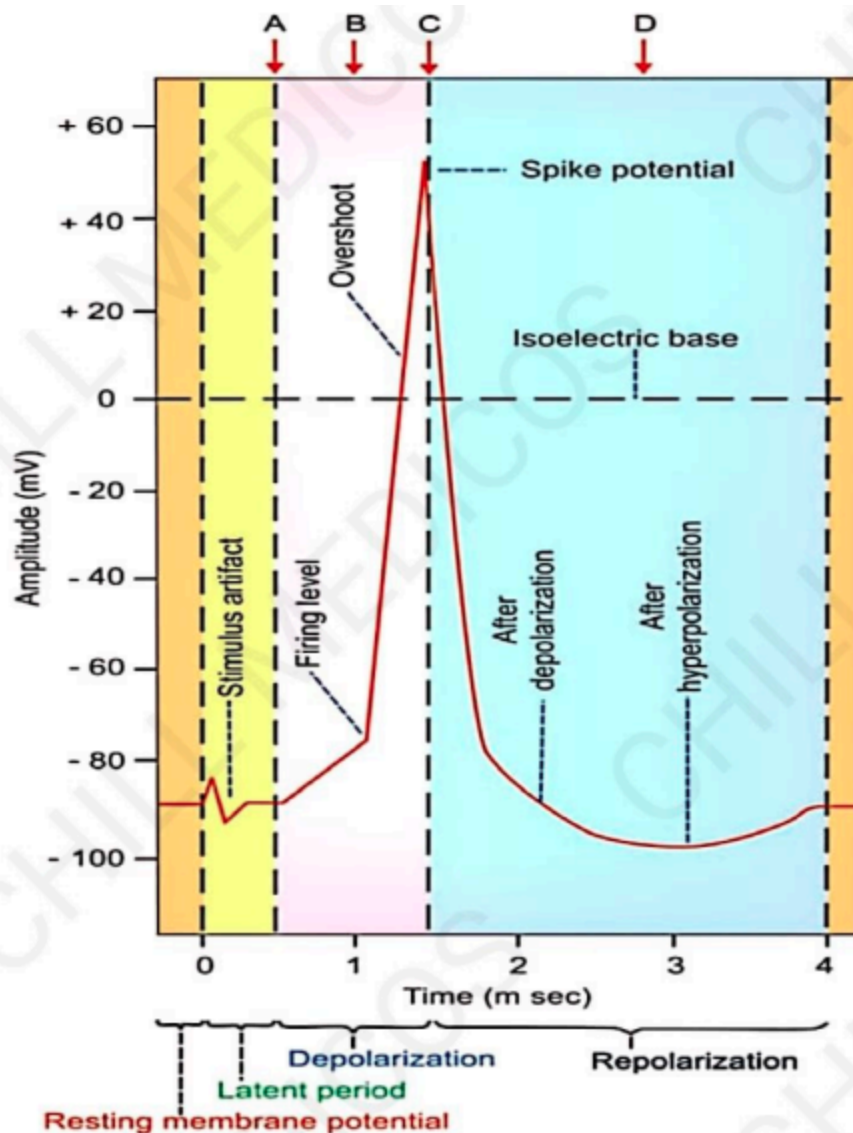
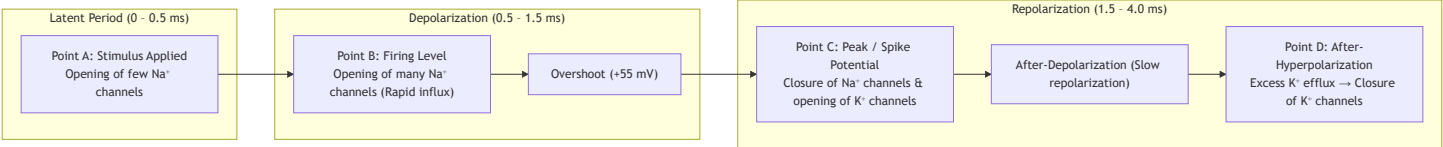


FIGURE 31.2: Action potential in a skeletal muscle

- A = Opening of few Na⁺ channels
- B = Opening of many Na⁺ channels
- C = Closure of Na⁺ channels and opening of K⁺ channels
- D = Closure of K⁺ channels

PHASES & CHANNEL DYNAMICS (TIMELINE)



IONIC BASIS OF ACTION POTENTIAL

During onset of
depolarisation



Voltage-gated Na^+
channels open (Slow influx
of Na^+)



Na^+ channel activation
(Depolarisation reaches 7
to 10 mV)



Na^+ channels start opening
at a faster rate
(Na^+ transport is short-
lived)



At the same time, K^+
channels start opening
(Leads to efflux of K^+ out
of cell)



Causing repolarisation
(K^+ channels open for
longer time)



Causes efflux of more

number of K^+ producing
more negativity inside



Causes hyperpolarisation



RMP restored